Received: 20 September 2024 | Accepted: 3 March 2025

DOI: 10.1111/mice.13461

RESEARCH ARTICLE

COMPUTER-AIDED CIVIL AND INFRASTRUCTURE ENGINEERING WILEY

A vision-based weigh-in-motion approach for vehicle load tracking and identification

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Funding information

National Research Foundation of Korea, Grant/Award Numbers: RS-2022-NR071667, RS-2024-00454760

Abstract

With the rapid increase in the number of vehicles, accurately identifying vehicle loads is crucial for maintaining and operating transportation infrastructure systems. Existing load identification methods typically rely on collecting vehicle load data from weigh-in-motion (WIM) systems when vehicles pass over them. However, cumbersome installation, high costs, and regular maintenance are the main obstacles that prevent WIM from being widely used in practice. This study introduces the visual WIM (V-WIM) framework, a vision-based approach for tracking and identifying moving loads. The V-WIM framework consists of two main components, the vehicle weight estimation and the vehicle tracking and location estimation. Vehicle weight is estimated using tire deformation parameters extracted from tire images through object detection and optical character recognition techniques. A deep learning-based YOLOv8 algorithm is employed as a vehicle detector, combined with the ByteTrack algorithm for tracking vehicle location. The vehicle weight and its corresponding location are then integrated to enable simultaneous vehicle weight estimation and tracking. The performance of the proposed framework was evaluated through two component validation tests and one on-site validation test, demonstrating its capability to overcome the limitations of existing methods.

1 | INTRODUCTION

The management and maintenance of transportation infrastructure are significantly impacted by the rapid increase in vehicle numbers. As a result, accurately identifying vehicle loads and their locations is a critical challenge for infrastructure managers (Oh et al., 2007; Salama et al., 2006). Overloaded vehicles pose a dual threat: They compromise traffic safety and degrade the structural integrity of roads and bridges, thereby shortening the lifespan of these structures and increasing maintenance requirements (Pais et al., 2013; Sianipar & Dowaki, 2014). A study conducted in Lokoja-Abuja, Nigeria, revealed that overloaded trucks reduced the service life of the pavement structure by approximately 10.48 years from its design life (O. O. Jacob et al., 2020). In Indonesia, statistics from the traffic police showed that, in 2019, out of 1,376,956 offenses, 136,470 vehicles were found to be over-dimensioned and overloaded. This phenomenon is a major cause of temporary changes in traffic speed and congestion (Sugari et al., 2022). A study in Poland examining the impact of overloaded vehicles on the whole life cycle cost of pavements demonstrated that reducing the percentage of overloaded vehicles from 23% to 5% resulted in an 11% decrease in the total life cycle cost borne by the road authority (Rys & Jaskula, 2019). These studies provide compelling evidence of the detrimental consequences of overloaded vehicles on transportation infrastructure. Therefore, accurately identifying both the vehicle load and its location is crucial for improving the safety and durability of transportation infrastructure.

Estimating the weight of vehicles accurately has long been a challenge for researchers. The conventional method of static weighing, which uses sensors and devices installed on the pavement, is considered reliable and widely used for weight restriction and overload enforcement around the world (B. Jacob & O'Brien, 2005; Karim et al., 2014). However, this method requires vehicles to divert to a weigh station and come to a complete stop, causing transportation interruptions. Additionally, the need for on-site staff makes it an uneconomical option (B. Jacob & Feypell-de La Beaumelle, 2010). To address these issues, the weigh-in-motion (WIM) system was developed (Cebon, 1990; Norman & Hopkins, 1952). This innovative technique allows for the weighing of heavy vehicles while in motion at normal or lower speeds, eliminating the need for diversions to weigh stations and enhancing its applicability, compared to static weighing. The most common type of WIM system, pavement WIM (P-WIM), utilizes a suite of sensors and devices, such as strip sensors, wheel or axle scale plates, bending plates, and piezoelectric cables, strategically placed beneath the pavement of the targeted lane to accurately measure the weight of vehicles (Sujon & Dai, 2021). While P-WIM has demonstrated superior convenience and effectiveness under real-world traffic conditions, its accuracy is highly dependent on the quality and quantity of sensors (Al-Qadi et al. 2016), making it difficult to implement widely due to its high cost. Another type of WIM system is bridge WIM (B-WIM), first proposed by Moses in 1979 (Moses, 1979). B-WIM systems use an instrumented bridge, with strain sensors positioned on the bridge to measure bending deformations. However, the prolonged passage of heavy vehicles over the bridge can lead to fatigue-induced deterioration or even structural damage, reducing the overall durability of the bridge (Sujon & Dai, 2021). In summary, static weighing systems require vehicles to divert to weighing stations, thereby causing traffic disruptions. P-WIM and B-WIM have been introduced to overcome these issues, but they still have limitations, such as high installation costs, high dependence on the quality of sensors, and the need for regular maintenance. Therefore, there is an increasing demand for a more robust and cost-effective WIM approach that is less dependent on equipment quality and maintenance.

The emergence of computer vision and deep learning algorithms has greatly improved the effectiveness of various techniques for inspecting and monitoring civil infrastructure. This includes advancements in structural displacement measurement (Y. Lee et al., 2022; Pan et al., 2023; Razavi et al., 2024; Wang et al., 2023, 2024; Nguyen et al., 2025), structural condition assessment (Perez-Ramirez et al., 2016; Rafiei & Adeli, 2017, 2018; Khatir et al., 2024), damage detection (Vy et al., 2024; Y. Lee et al., 2024), modal analysis and FE model updating technique (G. Park et al., 2023; J. Park et al., 2023; Y. Lee et al., 2023) and the broader area of structural health monitoring (Javadinasab Hormozabad et al., 2021; Pezeshki, Adeli, et al., 2023; Pezeshki, Pavlou, et al., 2023; Spencer et al., 2025). Computer vision-based methods have several advantages, including being low-cost, non-contact, and non-destructive, which makes them a promising alternative to conventional sensor-based methods. The early exploration of applying computer vision techniques to vehicle load identification began with image processing methods. To address the challenges associated with installing conventional WIM systems, the "nothing-on-road" concept was introduced. The novelty of this methodology lies in its capability to eliminate most sensors, devices, and cables, relying solely on cameras to measure the vehicle load. Ojio et al. (2016) presented a contactless B-WIM method, which uses cameras to measure both bridge deflections and vehicle axle spacings. Feng et al. (2020) also proposed a non-contact WIM technique, where the weight of vehicles was estimated by multiplying the tire–road contact pressure by the contact area, which was derived from tire deformation parameters. However, these approaches require manually determining the tire deformation parameters based on the number of pixels they occupy in the image, which can disrupt the process and affect result accuracy. Based on the Hertz contact theory (Johnson, 1987), Kong, Zhang, et al. (2022) developed a theoretical tire–road contact model for estimating vehicle weight in a non-contact manner, where the relationship between vertical force, tire deformation, and inflation pressure was established, providing the basis for vehicle weight identification. However, this study assumes that the contact area between the tire and the road surface follows a simple geometric shape, such as a rectangle or oval, which may limit the accuracy of the tire deformation calculations. To address this limitation, a comprehensive analysis was conducted using numerical 3D models of various tire configurations, demonstrating that the tire–road contact area has different shapes at varying levels of tire deformation. This led to the development of tire force-deformation equations (Kong, Wang, et al., 2022). Li et al. (2024) proposed a non-contact method for identifying overweight vehicles by using a vehicle vibration model and relevant specifications extracted through two deep learning models. However, the vehicle parameters used in this method are usually collected manually or empirically. Non-contact WIM techniques are a revolutionary solution in the field of vehicle weighing, as they address the issues of high cost and cumbersome sensor installation. However, both sensor-based WIM and non-contact WIM techniques can only estimate vehicle weight at specific locations where sensors or cameras are installed, limiting their ability to continuously monitor moving vehicles over an extended stretch of road or bridge.

In recent years, numerous vehicle tracking methods have been combined with WIM systems to address the problem of monitoring the location and weight of vehicles on traffic infrastructure systems. Zhicheng Chen et al. (2016) integrated weight data from a WIM system with locations of vehicles tracked using template-matching and partial-filtering techniques to obtain the spatial–temporal distribution of vehicle loads on a long-span bridge. Additionally, Dan et al. (2019) proposed a methodology for determining the transverse and longitudinal positions of vehicle loads on a bridge by combining information from a P-WIM system and multiple cameras. Despite the success of these methods in determining the weight and location of moving vehicles, one limitation is that the parameters of image processing methods must be manually adjusted by the user, which can significantly affect system performance. Therefore, addressing this limitation is crucial for enhancing the robustness and reliability of load identification systems in real-world applications.

Deep learning-based object tracking methods have shown great potential for improving the performance of traffic monitoring systems. Leveraging their high precision and adaptability to varying environmental conditions, B. Zhang et al. (2019) employed the deep learning-based detection model Faster region-based convolutional neural network (R-CNN) and a camera calibration technique to develop a non-contact system capable of capturing spatial–temporal information about the vehicles on bridges. Zhou et al. (2020) introduced a non-contact vehicle load identification approach, establishing a relationship between unique vehicle types and their corresponding weight information, while tracking vehicle positions using the Faster R-CNN model and the Kalman filter technique. Despite the effectiveness of the Faster R-CNN detector in vehicle detection within traffic videos, its two-stage detection process poses a computational limitation when compared to one-stage object detection models, such as those in the You Only Look One (YOLO) family. To address this limitation and ensure real-time monitoring capabilities, Ge et al. (2020) proposed a YOLOv3-based monitoring method. This method integrates vehicle weight information obtained from a WIM system with the position of the vehicle on the bridge, determined using the bounding box of both the vehicle and its tail, as predicted by the YOLOv3 model. Furthermore, Yang et al. (2022) introduced an automated method for generating fine-grained traffic load spectra by fusing the WIM value with vehicle spatial–temporal data obtained from the YOLOv3 detector and historical passing vehicle information. However, these methods remain dependent on the WIM system for vehicle load measurements, thereby inheriting its fundamental limitations.

Based on the analysis of the methods presented in the literature review, the following conclusions can be drawn:

1. Non-contact WIM methods based on computer vision have been proposed to address challenges related to the cost and sensor dependency of sensor-based WIM systems. However, these methods can only identify the weight of vehicles at fixed locations and are not capable of tracking the location of vehicles.
2. Vehicle tracking methods based on conventional image processing techniques combined with WIM systems have been introduced and can simultaneously determine the location and magnitude of the vehicle load. However, these methods require manual procedures and are susceptible to variations in environmental conditions.
3. Deep learning-based vehicle tracking methods such as Faster R-CNN and YOLOv3 are employed due to their high precision and generalization ability in object tracking tasks. The fusion of deep learning-based vehicle tracking methods with WIM systems has proven effective in producing stable tracking performance in different traffic scenarios and meeting the requirement of real-time traffic load monitoring. Nevertheless, these methodologies continue to rely on the WIM system for vehicle load measurements, thereby suffering from its inherent weaknesses associated with the WIM system that were discussed above.

To address the remaining challenges of the methods mentioned above, this paper proposes a visual WIM (V-WIM) framework that combines vision-based vehicle weight estimation and vision-based vehicle tracking to simultaneously identify the weight and location of vehicles on traffic infrastructure. In terms of vehicle weight estimation, the weight of the vehicle is determined based on the physical principle that the force between the tires and the road equals the product of the contact area and the contact pressure, which are obtained through computer vision techniques. For vehicle tracking, a real-time detection-based vehicle tracking is employed. By integrating the YOLOv8 detector (Josher et al., 2023) and ByteTrack tracker (Y. Zhang et al., 2022), the system continuously and accurately tracks the positions of multiple vehicles in complex traffic scenarios. This proposed vision-based method offers superior robustness and applicability compared to existing sensor-based methods, particularly in simultaneously monitoring the weight and location of vehicles. Additionally, its cost-effectiveness, non-contact nature, and ease of deployment make this method highly suitable for practical implementation in traffic conditions.

The main contributions of this paper are as follows:

1. Integrated vision-based vehicle load tracking and identification framework: This paper introduces a V-WIM method specifically designed to monitor both the magnitude and location of vehicle loads as they move on traffic infrastructure. To the best of our knowledge, this is the first attempt to integrate vision-based vehicle weight estimation and vision-based vehicle location estimation into a single framework. By leveraging computer vision techniques, the proposed method requires only visual data acquired from two cameras, eliminating the need for additional equipment such as weighing stations or sensors, which are required in existing methods.
2. Introducing a vision-based vehicle weight estimation approach: This study utilizes the relationship between tire–road contact force, contact pressure, and contact area to estimate vehicle weight. A novel approach combining computer vision and deep learning is proposed to extract tire deformation parameters from images. These parameters are then used to estimate the contact pressure and contact area. The proposed method demonstrates high accuracy and efficiency in estimating tire deformation parameters, compared to existing methods, resulting in precise vehicle weight estimation.

This paper is structured as follows: Section 2 introduces framework development in detail. Section 3 provides the framework validation. The discussion is presented in Section 4. Finally, Section 5 describes the conclusion.

2 | FRAMEWORK DEVELOPMENT

The proposed V-WIM framework, which uses visual information to identify the weight and location of vehicles, is composed of four main phases as shown in Figure 1: visual data collection and camera calibration, vehicle weight estimation, vehicle tracking and location estimation, and framework integration.

Phase 1 is visual data collection and camera calibration. In this phase, the visual information is collected from two videos, which will be used to estimate the weight and location of vehicles. The first video, captured by the weight estimation camera, will be used for vehicle weight estimation, and the second video, captured by the tracking camera, will be used for vehicle tracking and location estimation. Additionally, a camera calibration technique will be applied to the video to remove lens distortion, enhancing the precision of the vehicle tracking process. The detailed procedure will be explained in Section 2.1.

Phase 2 is the vehicle weight estimation. When a vehicle appears in the video captured by the weight estimation camera, a detection model will be used to detect crucial vehicle components such as tires, tire rims, and tire sidewall markings. Next, computer vision techniques will be utilized to extract tire deformation parameters (contact length, vertical deflection) and tire specifications (tire size, rim diameter, inflation pressure). Following this, the weight of the vehicle can be estimated by applying the physical principle that the tire–road contact force is equal to the product of contact area and contact pressure, which are obtained from the deformation parameters and tire specifications. Detailed information will be presented in Section 2.2.

Phase 3 is the vehicle tracking and location estimation. In this phase, vehicles will be tracked from the moment they appear in the video captured by the tracking camera. The location of each vehicle will be monitored continuously until it disappears from the video. Initially, the positions of moving vehicles in traffic videos are obtained through bounding boxes predicted by a detection model and a tracking algorithm. However, these positions are initially expressed in the image coordinate system. To rectify this, a coordinate transformation technique is used to convert tracked vehicle coordinates from the image coordinate system to the real-world coordinate system, providing actual locations of vehicles within the traffic infrastructure. The detailed procedure will be discussed in Section 2.3.

Phase 4 is framework integration. In this phase, two estimation processes from Phase 2 and Phase 3 (vehicle weight estimation, vehicle tracking, and location estimation) will be integrated into a single framework. A time synchronization process will be implemented to align the weight data with the corresponding positional data, enabling simultaneous vehicle weight estimation and tracking. The detailed process will be covered in Section 2.4.

2.1 | Visual data collection and camera calibration

The operational scenario for the V-WIM framework is illustrated in Figure 2. As a vehicle enters the estimation zone (indicated by the yellow line zone), it first passes

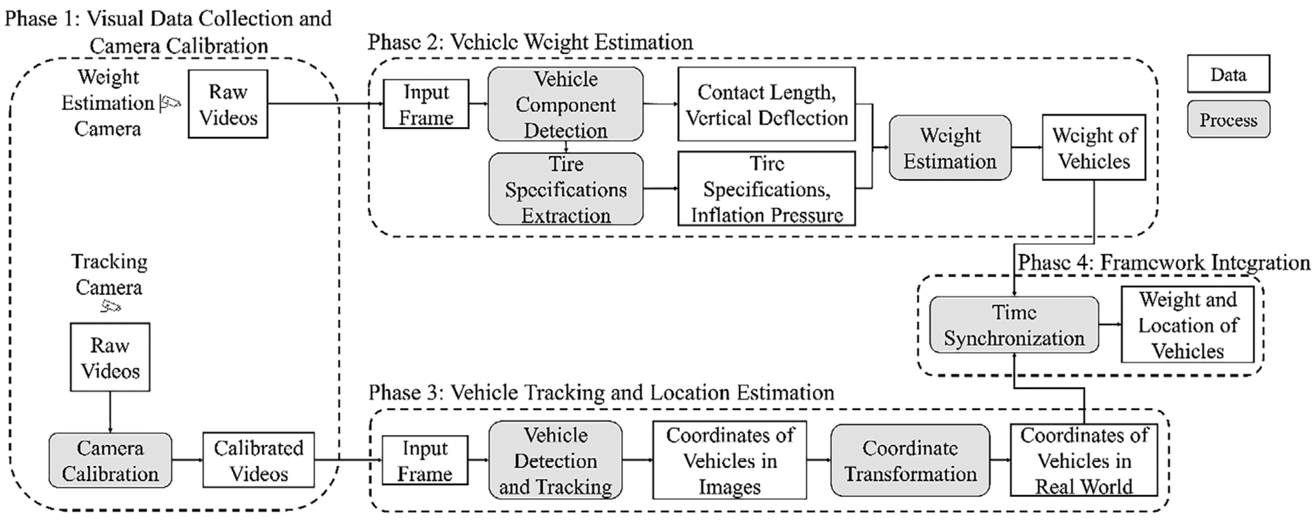

FIGURE 1 Flowchart of the proposed visual weigh-in-motion (V-WIM) framework.

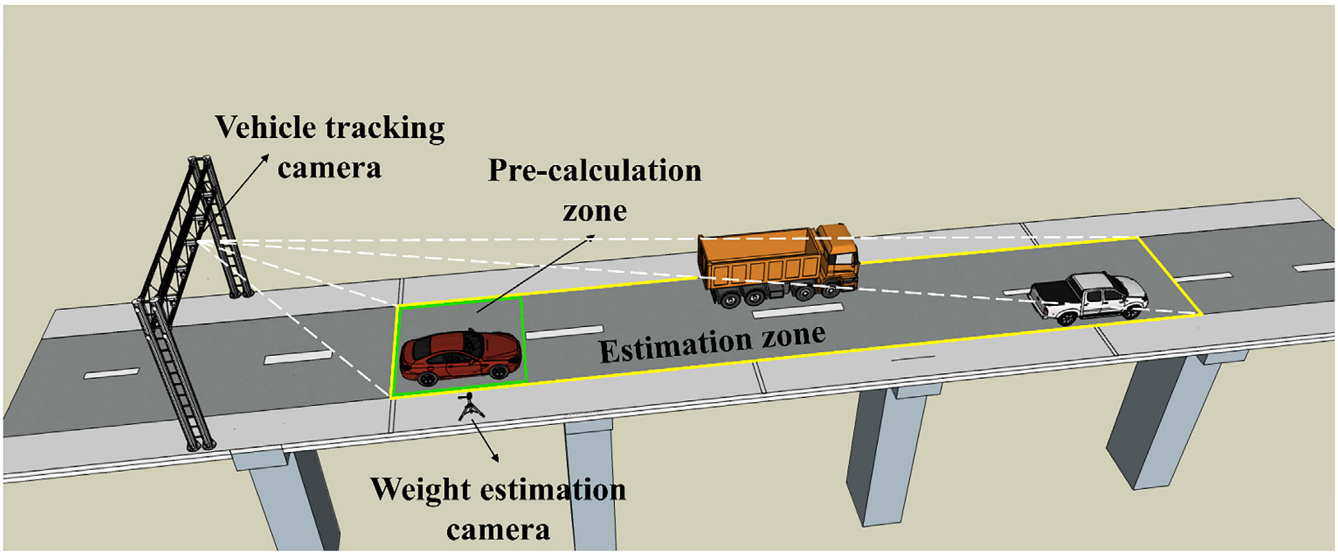

FIGURE 2 Overview of the proposed V-WIM framework.

through the pre-calculation zone (indicated by the green line zone). Within this initial zone, the weight of the vehicle is estimated by the vehicle weight estimation process. Concurrently, the location of the vehicle is determined by the vehicle tracking and location estimation process. These two components operate synchronously to integrate the estimated weight data with the corresponding positional data. This integrated approach allows the V-WIM framework to simultaneously identify the weight and the location of the vehicle as it moves through the estimation zone.

The proposed framework utilizes a dual-camera setup to monitor vehicles on a designated section of the traffic infrastructure. In the pre-calculation zone, the first camera, referred to as the weight estimation camera, is strategically placed on the roadside with its angle aligned perpendicular to the wheels of passing vehicles. This camera captures images of the tires to estimate vehicle weight. A second camera, named the vehicle tracking camera, is positioned at an elevated angle to monitor the traffic on the road. The video recorded from this camera will be used to detect and track vehicles within the estimation zone.

To enhance accuracy in tracking and locating moving vehicles in the estimation zone, a camera calibration technique needs to be applied to eliminate lens distortion from the vehicle tracking camera. It is noted that in the pre-calculation zone, the weight estimation camera is installed at an optimal location with its angle perpendicular to the vehicle wheels. This configuration minimizes measurement error (G. Lee et al., 2022), thereby eliminating the need for camera calibration in this context.

The procedure of camera calibration involves estimating the intrinsic parameters of a camera, including focal length, optical center (the principal point), skew coefficient, and lens distortion coefficient. Within the scope of this research, the methodology proposed by Z. Zhang (2000) was used to remove radial distortion. A brief overview of the camera calibration procedure is presented as follows:

Given the ith 3D point $[X_i Y_i Z_i]$ in the world coordinates system and corresponding 2D point $[u_i v_i]$ in the image coordinates system, the 3D point is projected onto the image as a 2D point by using the extrinsic matrix (rotation matrix and translation vector) and intrinsic matrix.

$$s \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = [\boldsymbol{R}|\boldsymbol{t}]\boldsymbol{K} \begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} \tag{1}$$

where $\boldsymbol{R}$ represents a 3 by 3 rotation matrix, $\boldsymbol{t}$ is the 1 by 3 translation vector, and $\boldsymbol{K}$ denotes the camera intrinsic matrix, which is expressed as

$$\boldsymbol{K} = \begin{bmatrix} f.m_x & \gamma & c_x \\ 0 & f.m_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

In this context, $[c_x c_y]$ represents the optical center, commonly known as the principal point, while f denotes the focal length. The parameters m_x and m_y are the scale factors in the x and y directions, respectively, and γ represents the skew coefficient. The intrinsic matrix can be determined by having the camera observe a planar pattern at a few (at least two) different orientations. Once the camera calibration is completed, lens distortions in image frames will be eliminated, thereby enhancing the effectiveness of the vehicle tracking process.

2.2 | Vehicle weight estimation

In this section, a computer vision-based vehicle weighing approach is proposed for estimating the weight of vehicles. Figure 3 shows the camera configuration for the vehicle weight estimation. As described in the overview of the proposed framework, vehicle weight estimation occurs while the vehicle is in the pre-calculation zone. In this zone, a weight estimation camera is used to capture tire images of the passing vehicle, which are then used as input data for vehicle weight estimation. The flowchart for the vehicle weight estimation is described in Figure 4.

The first step is the tire component detection. This process employs a tire component detection model, trained using the YOLOv8 detection framework, which is enhanced through transfer learning techniques to detect tires, rims, and sidewall markings from raw tire images.

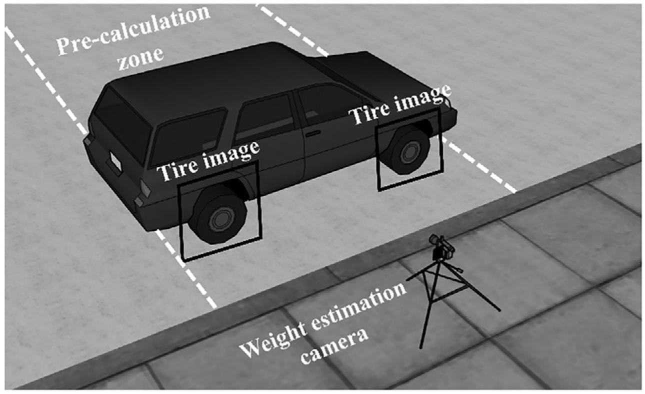

FIGURE 3 Camera configuration of the vehicle weight estimation.

The second step is the tire specification extraction. The optical character recognition (OCR) technique is applied to extract tire specifications from the sidewall markings detected in the previous step. This process enables the determination of critical parameters such as tire size, rim diameter, and tire–road contact pressure.

The third step is the tire–road contact area estimation. In this step, the tire–road contact length and vertical deflection are estimated from the detected tire. The rim diameter is then used to compute the scale factor necessary for converting measurements from pixels to physical units. Ultimately, the tire–road contact area is calculated using the estimated tire–road contact length, vertical deflection, and tire size.

The final step is the vehicle weight estimation. The weight of the vehicle is determined based on the physical principle that the force between the tires and the road is equal to the product of the contact area and contact pressure.

2.2.1 | Tire component detection

In the V-WIM framework, the raw tire images captured by the weight estimation camera lack detailed information, making it infeasible to directly obtain the contact area and contact pressure, which are critical attributes for estimating the weight of vehicles. Therefore, a series of preprocessing steps is necessary to extract useful parameters from the raw images. A crucial preprocessing step involves detecting tires, rims, and sidewall markings in the captured images.

To develop a detection model for detecting tires, rims, and sidewall markings, the first step was to build a tire image dataset. Tire images were collected at Chungbuk National University with permission from the vehicle owners. The collected dataset consists of 600 images from several popular vehicle types, including passenger cars

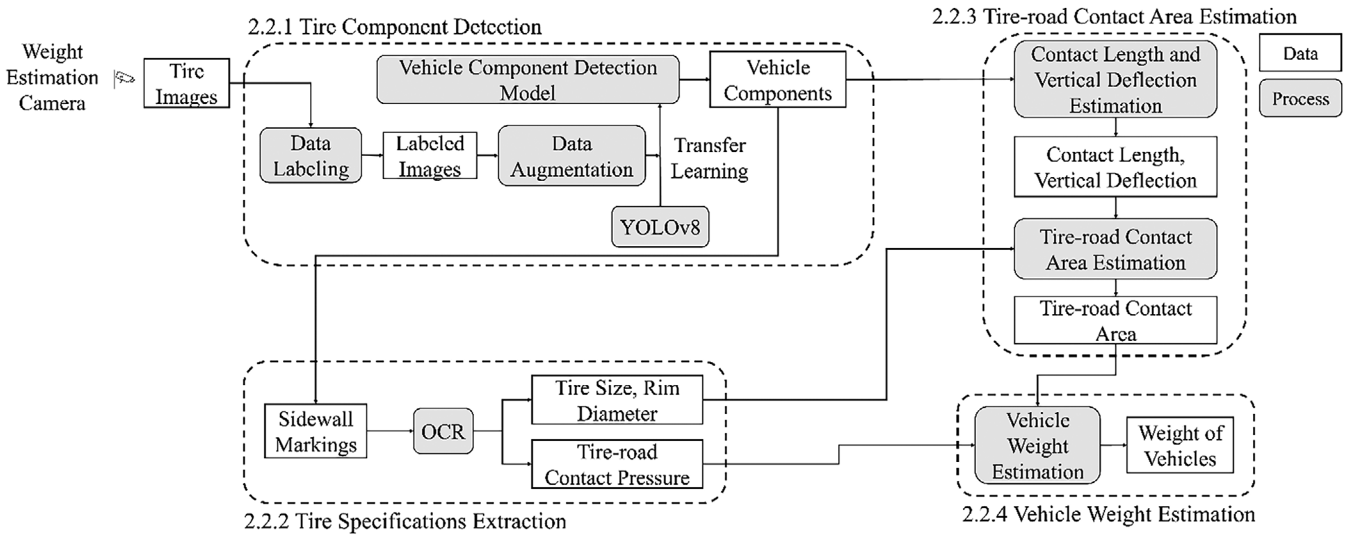

FIGURE 4 Flowchart of the vehicle weight estimation.

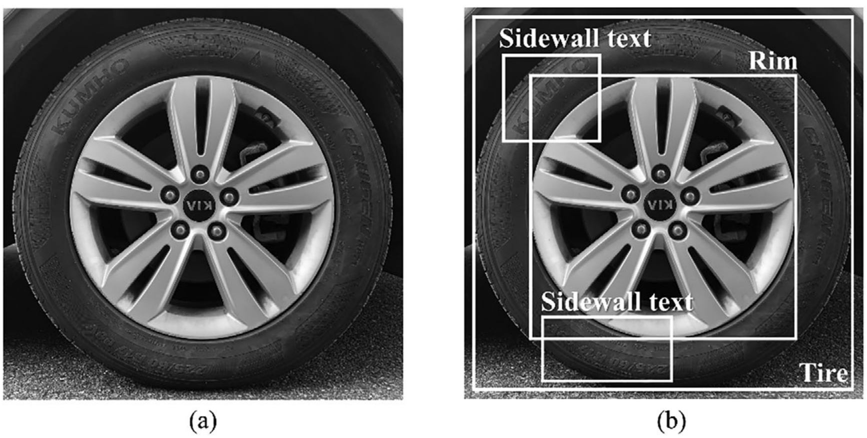

FIGURE 5 Visualization for labeling the image: (a) the original image; (b) the labeled image.

(sedans, SUVs) and light trucks, as well as various tire sizes and brands. Each image is labeled with three classes: tire, rim, and sidewall text, which correspond to three distinct components, including the tire, rim, and sidewall markings. The original image and its corresponding labels are depicted in Figure 5a,b, respectively. Next, the tire image dataset is first split into two subsets, a training set and a validation set, with a ratio of 7:3. Then, the training set is augmented through augmentation techniques including adjusting brightness, blurring, and adding noise to enhance dataset diversity. Image augmentation not only helps reduce overfitting but also simulates the influence of extreme weather conditions commonly encountered in real-world scenarios. For example, blurring and adding noise can simulate fog, snow, or light rain, whereas adjusting brightness can simulate low-light conditions. In contrast, the validation set remains unchanged to ensure objective evaluation. Table 1 represents the distribution of images in the dataset, and Figure 6 shows the diversity of the tire images with different augmentation techniques.

TABLE 1 A detailed distribution of the tire dataset.

	Original images	Augmented images	Total
Training set	420	840	1260
Validation set	180	0	180

The learning curve of the training process is shown in Figure 7. The three columns on the left show both the training and validation losses for each type of loss function. The two columns on the right present evaluation metrics, including precision, recall, mean average precision at a threshold of 0.50 (mAP50), and mean average precision across thresholds from 0.50 to 0.95 (mAP50-95) on the validation set. Except for the validation set, the classification loss shows slight fluctuations. In general, there is consistency in loss values in both the training and validation sets, which implies there is no overfitting. Moreover, high precision, recall, mAP50, and mAP50-95 metrics of the

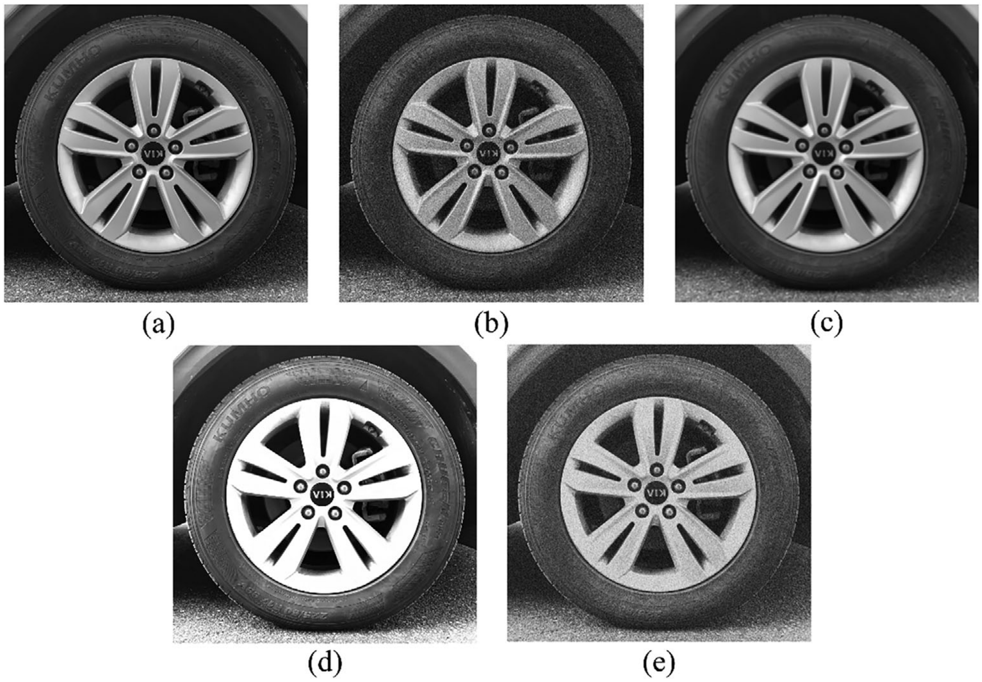

FIGURE 6 Diversity of tire images after applying image augmentation techniques: (a) the original image; (b) the image with noise; (c) the blurred image; (d) the image with increased brightness; and (e) the image with noise, blur, and decreased brightness.

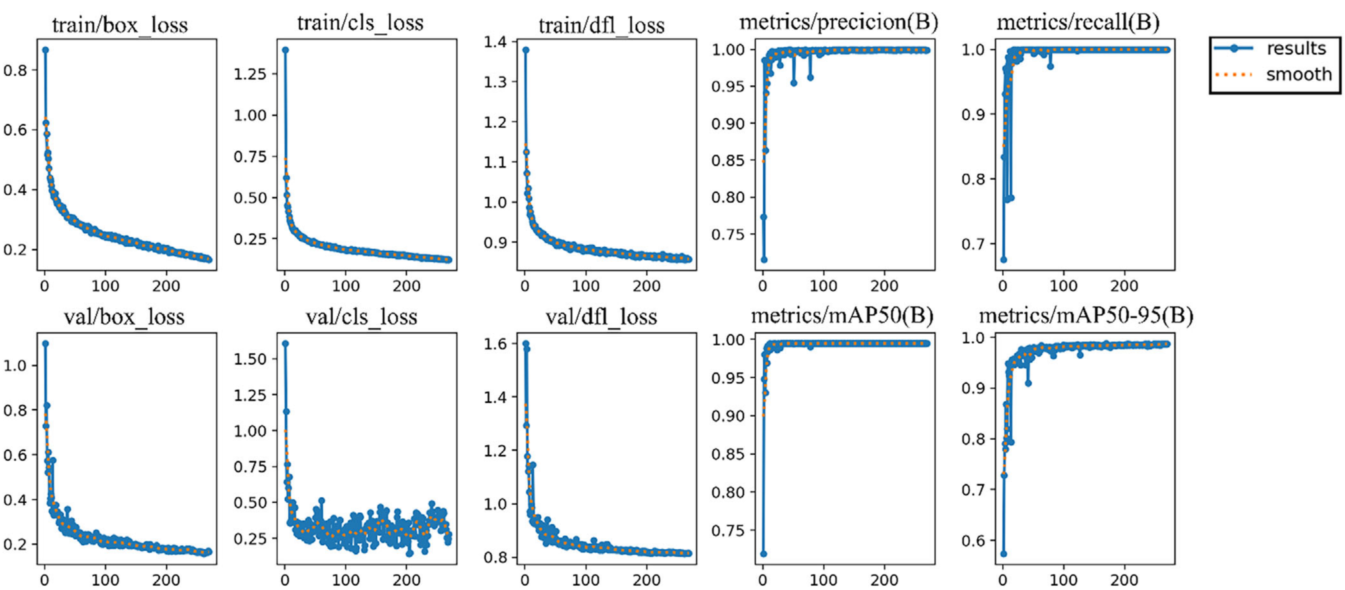

FIGURE 7 Illustration of the learning curve for the tire, rim, and sidewall text detection model.

validation set demonstrated good performance in detecting tires, rims, and sidewall text.

2.2.2 | Tire specifications extraction

From the sidewall text detected by the tire component detection model in Section 2.2.1, an OCR technique is utilized to extract specifications, such as the tire brand, tire model, tire section width, tire aspect ratio, rim diameter, load index, speed rating, etc. These technical parameters are essential for estimating the tire–road contact area and contact pressure. The tire technical parameters are illustrated in Figure 8. However, challenges emerge due to the curved form of the sidewall text combined with the similarity in color between the characters and the tire surface,

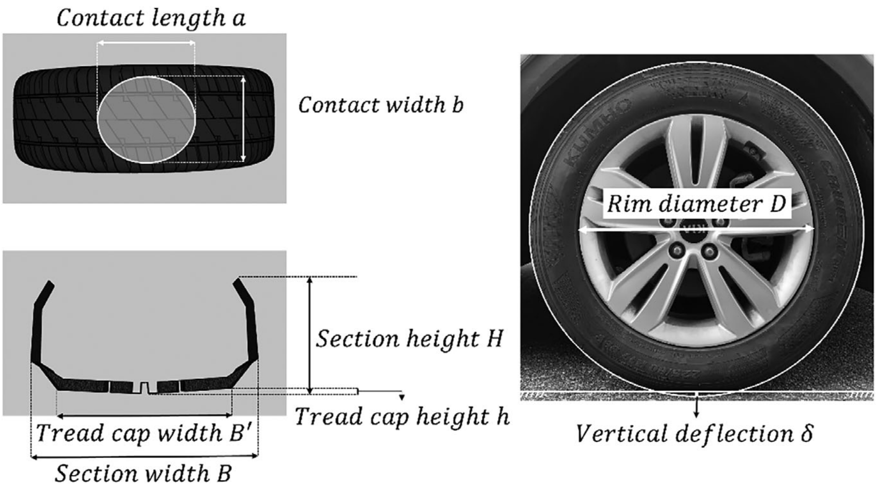

FIGURE 8 Illustration of the tire technical parameters.

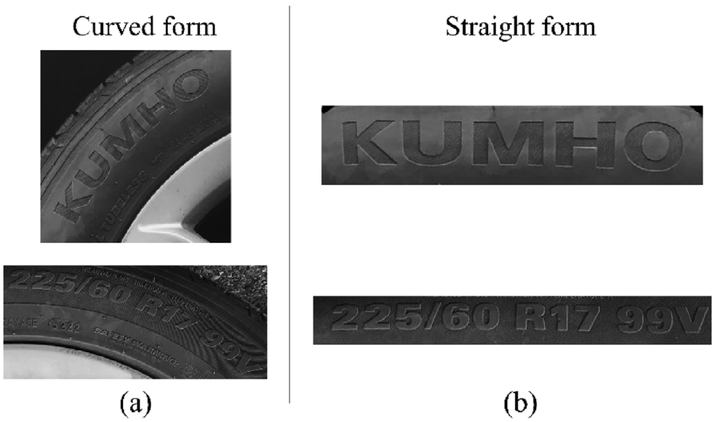

FIGURE 9 Illustration of curved text rectification: (a) curved form and (b) straight form.

as presented in Figure 9a, resulting in a reduction in character recognition performance. To mitigate these issues, image preprocessing steps were implemented to enhance the readability of the sidewall text. Initially, a curved text rectification method (frotms, 2020) is employed to rectify the curved sidewall text into a straight format as shown in Figure 9b.

Next, an image binarization technique based on Otsu's algorithm is applied to segment the image into foreground and background regions by determining an optimal threshold value. This threshold value distinguishes pixels belonging to the foreground (characters) from those constituting the background (tire surface). However, the binarization method produces suboptimal results, leaving residual background noise, as shown in Figure 10b. To address this issue, a morphology operation is performed to remove connected components in the binary image that contain fewer pixels than a predefined threshold. After applying these preprocessing steps, the readability of the sidewall text is significantly improved, leading to enhanced recognition performance. This enhancement is evident in the result presented in Figure 10c.

To recognize the characters in the preprocessed sidewall text, an open-source OCR is employed (rkcosmos, 2024). The recognition results are presented in Figure 11. Several useful pieces of information can be obtained from tire specifications (225/60 R17 99V). Specifically, the first three digits identify the tire section width B expressed in millimeters, in this case, 225 mm. From the tire section width B, the tire tread cap with B' can be determined based on the ratio of the tire tread cap with B' to the tire section width B. This ratio ranges from 0.60 to 0.90, where a mean ratio of 0.75 is accepted (Feng et al., 2020; Kong, Wang, et al., 2022). Thus, the tire tread cap width B' can be calculated as 168.75 mm ($225 \times 75\% = 168.75$). The next two digits denote the tire aspect ratio, reflecting the ratio of tire section height to section width, here specified as 60%. Consequently, the tire section height H can be derived as 135 mm ($225 \times 60\% = 135$). Following this, an alphabetic character signifies the tire internal construction, such as "R," indicating radial construction. Subsequently, a numerical value indicates the tire rim diameter D in inches, in this case, 17 in. (431.8 mm). Finally, the concluding alphanumeric combination indicates the tire service description, including its load index (99) and speed rating (V).

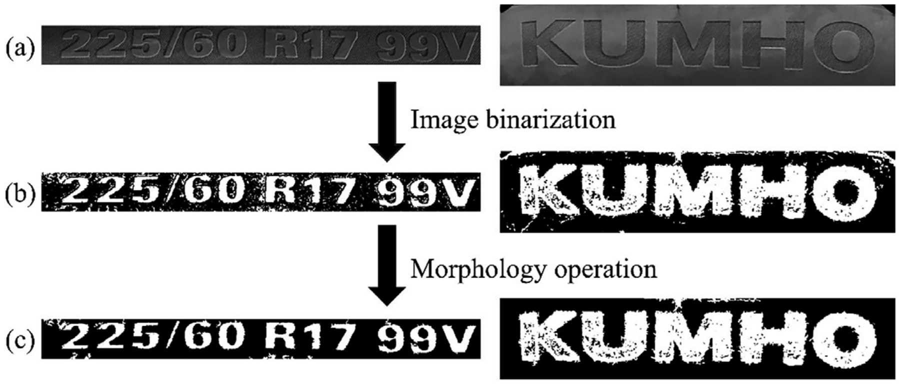

FIGURE 10 Process for separating characters from the tire surface: (a) the original image, (b) the image after segmentation, and (c) the enhanced sidewall text.

FIGURE 11 The results of character recognition in the sidewall text.

By using the OCR technique and standard design ratios, the essential parameters for estimating the tire–road contact area, including the tire tread cap with B', the section height H, and the tire rim diameter D, are identified. The remaining parameters, such as the tire internal construction, load index, and speed rating are ignored due to their non-relevance within this context.

In terms of tire–road contact pressure, this study follows an assumption that the tire–road contact pressure and the tire inflation pressure of a tire are identical at the same loading condition (Yap, 1989). In the context of this study, the manufacturer-recommended tire inflation pressure value can be used as an alternative for the actual tire pressure. This value is obtained from the tire technical database provided by the manufacturer, based on tire specifications extracted from the sidewall text. However, it is acknowledged that there is always a difference between the actual and manufacturer-recommended inflation pressure. To address this problem, several experiments have been conducted to investigate the relationship between these two values (Zhao Chen et al., 2018; Mohsenimanesh et al., 2009) and revealed that, with 95% confidence, the ratio of actual inflation pressure to recommended inflation pressure is 1.05–1.15 for light trucks and 1.20–1.30 for heavy trucks. In previous works, a mean ratio of 1.10 for light trucks and 1.25 for heavy trucks was accepted (Kong, Wang, et al., 2022; Kong, Zhang, et al., 2022). Table 2 presents the extracted tire specifications from the sidewall text.

2.2.3 | Tire–road contact area estimation

The tire–road contact area is one of the main parameters for estimating the weight of vehicles. Calculating the contact area is usually based on the assumption that it has a geometric shape, such as rectangular or oval, an assumption that is used in some studies (Feng et al., 2020; Kong, Zhang, et al., 2022). In addition, the relationship between the contact area and tire vertical deflection is investigated (Kong, Wang, et al., 2022), and it is demonstrated that the contact area has different shapes at different tire vertical deflection levels. In this paper, both types of methods, including geometric-based and deflection-based methods, are applied to measure the tire–road contact area. The equations of all the methods are as follows:

Rectangular method:

$$S = ab \tag{6}$$

Oval method:

$$S = \frac{8+\pi}{12}ab \tag{7}$$

TABLE 2 The tire technical parameters extracted by optical character recognition.

Brand	Identifier	Section width	Tread cap width	Aspect Ratio	Section height	Rim diameter	Inflation pressure
KUMHO	225/60 R17 90V	225 mm	168.75 mm	60	135 mm	17 in	51 psi

Deflection method (Kong, Wang, et al., 2022):
For cars:

$$S = \begin{cases} 2\delta\sqrt{D+2H-\delta}\sqrt{h+0.25\dfrac{B'^2}{h}}-\delta, & b < B' \\ 1.5B'\sqrt{(D+2H)\,\delta-\delta^2}, b = B' \end{cases} \quad (8)$$

For trucks:

$$S = \begin{cases} 0.6\pi\delta\,(D+2H-\delta)\,, & b < B' \\ B'\left[2\sqrt{(D+2H)\,\delta-\delta^2}-(1-0.25\pi)\,B'\right], b = B' \end{cases} \quad (9)$$

where S is the tire–road contact area, a and b represent the contact length and the contact width, respectively, δ is the tire vertical deflection, D denotes the tire rim diameter, B' is the tire tread cap width, and H is the tire section height, h is the tread cap height of the tire.

In practice, accurately determining the tire contact width b for a moving vehicle poses significant challenges. Therefore, within the context of this study, an assumption is that the tire contact width b is equal to the tire tread cap width B' ($b = B'$) is applied (Feng et al., 2020). Based on this assumption and Equations (6)–(9), the estimation of the contact area relies solely on the tire contact length a, contact width b, tire vertical deflection δ, tire rim diameter D, and tire section height H. Notably, the tire tread cap width B' (contact width b), tire rim diameter D, and tire section height H are identified using the OCR technique as described in Section 2.2.2. Therefore, the remaining parameters needed to estimate the tire–road contact area are the contact length a and the tire vertical deflection δ. The detailed estimation process of these two parameters will be presented below.

To the best of our knowledge, two main methods have been used in the literature to obtain the contact length a and the tire vertical deflection δ, which are image processing-based (Feng et al., 2020; Kong, Zhang, et al., 2022) and deep learning-based methods (J. Zhang et al., 2023). Both approaches generally aim to separate the tire from the background to facilitate the estimation of contact length and vertical deflection. In the image processing-based approach, a series of techniques, such as image segmentation/image thresholding, and morphology operation (dilation/erosion) was applied to separate the tire from the background. However, the inherent weaknesses of these image processing methods include their susceptibility to background variations, leading to unstable performance. In contrast, deep learning-based methods, particularly semantic segmentation, have demonstrated more robust performance in segmenting the tire and background, compared to image processing-based methods. Nonetheless, data labeling for semantic segmentation tasks is time-consuming and labor-intensive, as objects in images must be accurately labeled at the pixel level.

To overcome limitations observed in previous work, this paper proposes a novel and automated approach for estimating both the tire–road contact length a and the tire vertical deflection δ. Our proposed method leverages the predicted bounding box of the tire obtained from the tire component detection model as presented in Section 2.2.1. By employing a deep learning model for tire detection, our approach ensures the stability and robustness of the predicted results. Furthermore, the tire is labeled with a rectangular shape, requiring only two points (top-left and bottom-right), thereby mitigating the time-consuming and labor-intensive problem of image data labeling.

The procedure of the proposed method is delineated as follows:

Initially, a circle is generated with its center point O positioned at the centroid of the bounding box. The radius of this circle is defined as half the width of the bounding box. However, due to vertical deformation, the height of the bounding box is smaller than its width, causing the center point O to be higher than the actual tire center along the y-coordinate, resulting in the generated circle not fitting to the tire edge, as shown in Figure 12a. To rectify this misalignment, the y-coordinate of the center point O is adjusted downward by a distance equal to the difference between the width and height of the bounding box. After this adjustment, the new circle fits to the tire edge (except for the lower half of the tire) as illustrated in Figure 12b.

Subsequently, the tire–road contact length a is determined through the two intersection points between the bottom edge of the bounding box and the lower half of the circle ($A1$, $A2$) as shown in Figure 13a. Following this, the tire vertical deflection δ is also estimated. From the identified intersection points ($A1$, $A2$), a central point $D1$ is created such that $A1D1 = A2D1$. The vertical deflection of the tire is then calculated as the distance between the central point $D1$ and the lowest point on the circle $D2$ as depicted in Figure 13b.

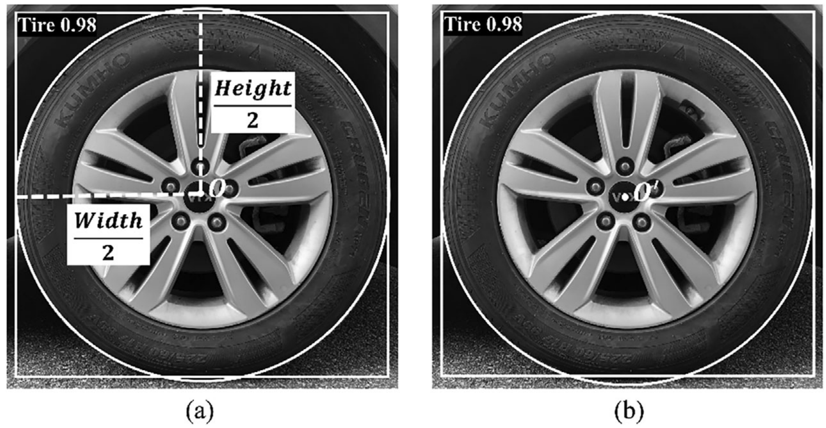

FIGURE 12 Process of creating and adjusting a circle around the tire: (a) the original circle and (b) the circle after adjustment.

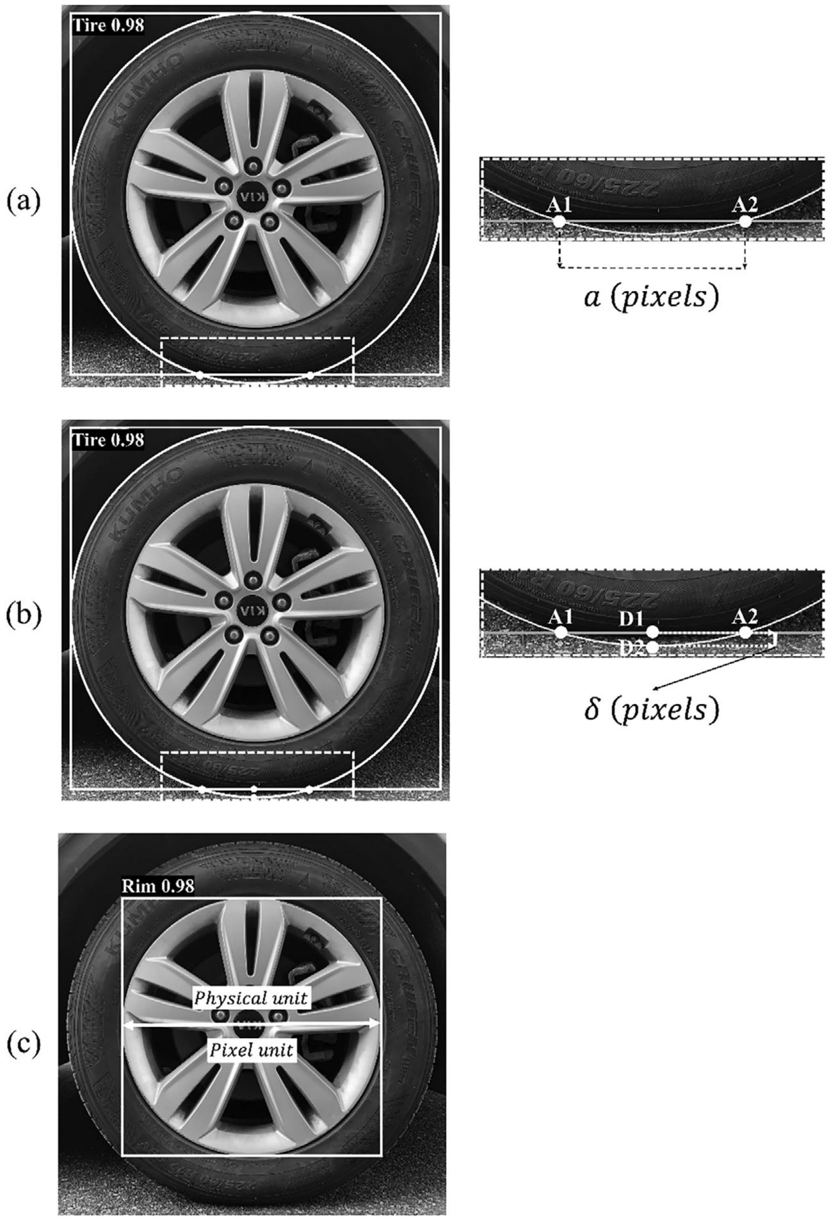

FIGURE 13 Illustration of estimating the tire–road contact length, tire vertical deflection, and scale factor: (a) the tire–road contact length, (b) the tire vertical deflection, and (c) the scale factor.

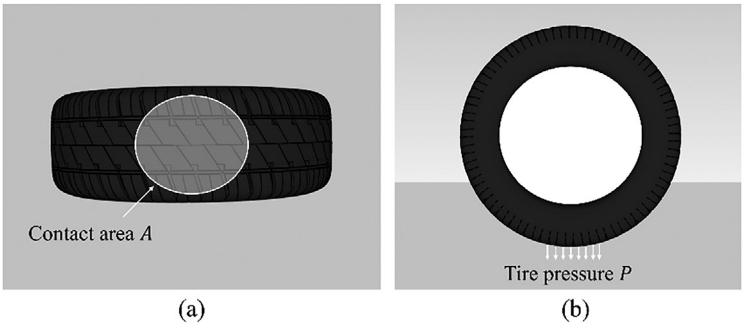

FIGURE 14 Illustration of the tire–road contact area and the contact pressure: (a) tire–road contact area and (b) tire pressure.

It is important to note that the estimated values of both the tire–road contact length a and the tire vertical deflection δ are currently expressed in pixel units. Therefore, establishing a scale factor to convert these dimensions from pixel units to physical units (e.g., mm) is necessary. The scale factor is determined by dividing the known dimension in physical units by its corresponding dimension in pixel units. In this paper, the dimensions of the rim are used to calculate the scale factor. The dimension of the rim in pixel units is obtained from the width or height of the predicted bounding box of the rim. The physical dimension (mm) is derived from the rim diameter extracted from the sidewall markings as detailed in Section 2.2.2. Figure 13c illustrates the estimation of the scale factor. In this case, the physical dimension and pixel dimension of the rim are 17 in. (431.8 mm) and 1714 pixels, respectively.

Based on the contact width b, the tire rim diameter D, and the tire section height H extracted from the sidewall markings via the OCR technique, along with the tire contact length a, tire vertical deflection δ estimated automatically through the predicted bounding box of the tire as described previously, the contact area between the tire and the road surface can now be determined.

2.2.4 | Vehicle weight estimation

To estimate the weight of a vehicle, a fundamental physics principle is applied, that is, the tire force F_t between the tire and the road is equal to the product of tire–road contact pressure P and contact area A (Feng et al., 2020; Kong, Wang, et al., 2022) as depicted in Figure 14. In other words, the weight of the vehicle can be inferred by summing the contact forces exerted by all tires as expressed in the following equation:

$$W = \sum_{i=1}^{N} F_{t_i} = \sum_{i=1}^{N} P_i A_i \tag{10}$$

where W is the total weight of the vehicle, N is the number of tires, F_{t_i} the tire–road contact force when the ith tire exerted onto the road surface, P_i and A_i are the tire–road contact pressure and the tire–road contact area of the ith tire, respectively.

Estimating the weight of the vehicle requires estimating both the contact pressure and contact area for each tire. The contact pressure can be inferred from the tire inflation pressure, as discussed in Section 2.2.2, while the contact area can be determined using the methods described in Section 2.2.3.

2.3 | Vehicle tracking and location estimation

The tracking and localization of moving vehicles within the estimation zone are essential parts of the proposed method, which is designed to track and locate vehicles through images captured by a tracking camera. The vehicles are first detected by the detection model, then the detected vehicles are used as input for a tracking algorithm, which is responsible for tracking vehicles continuously and consistently. Furthermore, the positions of tracked vehicles in the image coordinate system are transformed into the real-world coordinate system. Based on this, the trajectory of vehicles moving within the estimation zone can be identified. The flowchart for the vehicle tracking and location estimation is presented in Figure 15.

2.3.1 | Vehicle tracking

In this paper, YOLOv8 is used as a vehicle detection model due to its high accuracy and real-time processing capabilities, compared to other two-stage detection models such as R-CNN (Girshick et al., 2014) and Faster R-CNN (Ren et al., 2017). Additionally, the ByteTrack tracker is integrated with YOLOv8 to track vehicles by assigning a unique identifier (ID) to each vehicle. The combination of YOLOv8 and ByteTrack not only ensures a real-time vehicle detection and tracking but also provides accurate and consistent IDs for vehicles. This is crucial in this study because vehicle

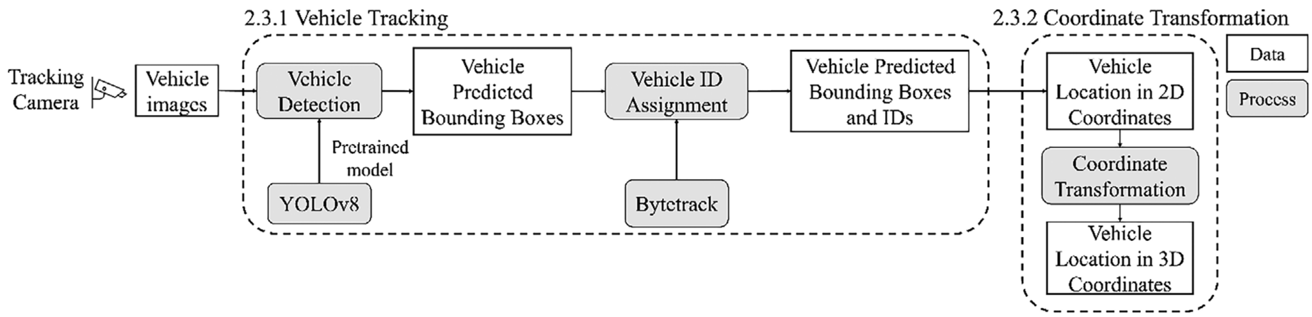

FIGURE 15 Flowchart for the vehicle tracking and location estimation.

weights are associated with their respective vehicles based on their IDs during the monitoring process.

The backbone network of YOLOv8 is built on a modified CPSDarknet53 architecture, which includes CBS, C2f, and SPPF modules to enhance feature extraction and reduce inference time. The neck incorporates a feature pyramid network (Lin et al., 2017) and path aggregation network (Liu et al., 2018) for effective multi-scale feature fusion. The head employs a dual-branch design for simultaneous object classification and bounding box regression.

Since YOLOv8 is a standalone detection model, it cannot accurately track multiple objects frame by frame, which can lead to missed detections during occlusions. Therefore, ByteTrack algorithm is utilized. ByteTrack employs a two-stage association process to track multiple vehicles by assigning a unique ID to each vehicle. In the first stage, high-confidence predicted bounding boxes are associated, while the second stage deals with low-confidence predicted bounding boxes. This approach differs from previous methods (Bewley et al., 2016; Liang et al., 2022), as they consider only high-confidence predictions. By associating all predicted bounding boxes, ByteTrack minimizes the exclusion of low-confidence predictions, which may occur due to obscured or distant objects. Consequently, this advanced association strategy enhances vehicle tracking performance, particularly in complex traffic scenarios.

2.3.2 | Coordinate transformation

With the integration of YOLOv8 and ByteTrack, moving vehicles in the traffic videos can be tracked. However, the positions of tracked vehicles are in the pixel coordinate system, which is not suitable for practical traffic monitoring since real-life applications need to identify vehicle positions in the real-world coordinate system. Therefore, a coordinate transformation process must be conducted to convert the 2D pixel coordinates of tracked vehicles in the image space into 3D coordinates in the real-world space. To achieve this, three coordinate systems have been established: the 2D image coordinate system, the 3D camera coordinate system, and the 3D world coordinate system, as shown in Figure 16.

Denote a 2D point p within the image coordinate system as $p = (u, v)$ and its corresponding 3D point P within the real-world coordinate system as $P = (X_w, Y_w, Z_w)$. And point P is also denoted as $P = (X_c, Y_c, Z_c)$ in the camera coordinate system. According to the previous works (Faugeras, 1993; Forsyth & Ponce, 2002), the relationship among the 2D point p, the 3D point P, and the coordinate transformation matrix M is defined as in Equation (12):

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \boldsymbol{M} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{11}$$

where M is defined as

$$M = [R|T] \times K \tag{12}$$

Here, the $[R|T]$ is the extrinsic matrix and is responsible for transforming the 3D world coordinate system to the 3D camera coordinate system, K is the intrinsic matrix and is used to transform the 3D camera coordinate system to the 2D image coordinate system, and s is an arbitrary scale factor.

With an expansion of the matrix M, Equation (11) can be rewritten as

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{13}$$

Notably, in the context of the V-WIM framework, the vehicles move on the road surface, which is usually an approximate plane. Consequently, all vehicle positions on this plane share an identical height coordinate Z_w, which

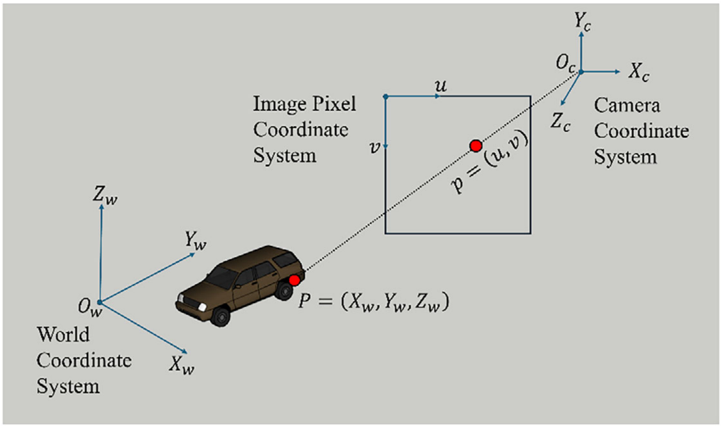

FIGURE 16 Three coordinate systems in the coordinate transformation process.

is assumed to be 0. Hence, Equation (13) turns into:

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix}\begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} \tag{14}$$

Equation (14) may be expressed as

$$u - \frac{X_w m_{11} + Y_w m_{12} + m_{13}}{X_w m_{31} + Y_w m_{32} + m_{33}} = 0 \tag{15}$$

and

$$v - \frac{X_w m_{21} + Y_w m_{22} + m_{23}}{X_w m_{31} + Y_w m_{32} + m_{33}} = 0 \tag{16}$$

Combine Equations (15) and (16):

$$\begin{cases} X_w m_{11} + Y_w m_{12} + m_{13} - uX_w m_{31} - uY_w\, m_{32} = um_{33} \\ X_w m_{21} + Y_w m_{22} + m_{23} - vX_w m_{31} - vY_w\, m_{32} = vm_{33} \end{cases} \tag{17}$$

If there are N reference points in both the 3D world coordinate system $(X_{wi}, Y_{wi})(i = 1, 2, \ldots, n)$ with their corresponding 2D coordinate system $(u_i, v_i)(i = 1, 2, \ldots, n)$ are known, leads to $2N$ linear equations. Those equations can be rewritten in matrix form as follows:

$$\begin{bmatrix} X_{w1} & Y_{w1} & 1 & 0 & 0 & 0 & -u_1X_{w1} & -u_1Y_{w1} \\ 0 & 0 & 0 & X_{w1} & Y_{w1} & 1 & -v_1X_{w1} & -v_1Y_{w1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{wi} & Y_{wi} & 1 & 0 & 0 & 0 & -u_iX_{wi} & -u_iY_{wi} \\ 0 & 0 & 0 & X_{wi} & Y_{wi} & 1 & -v_iX_{wi} & -v_iY_{wi} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{wn} & Y_{wn} & 1 & 0 & 0 & 0 & -u_nX_{wn} & -u_nY_{wn} \\ 0 & 0 & 0 & X_{wn} & Y_{wn} & 1 & -v_nX_{wn} & -v_nY_{wn} \end{bmatrix}$$

$$\times \begin{bmatrix} m_{11} \\ m_{12} \\ m_{13} \\ m_{21} \\ m_{22} \\ m_{23} \\ m_{31} \\ m_{32} \end{bmatrix} = m_{33}\begin{bmatrix} u_1 \\ v_1 \\ \vdots \\ u_i \\ v_i \\ \vdots \\ u_n \\ v_n \end{bmatrix} \tag{18}$$

Assuming $m_{33} = 1$, the unknown elements m_{ij} of the coordinate transformation matrix M can be determined by applying the least squares method under the conditions $N \geq 6$. Once the coordinate transformation matrix is obtained, the position of the moving vehicle on the road surface can be calculated based on its pixel coordinates in the image. From that, the vehicle trajectory can be identified by performing this operation to each image frame.

2.4 | Framework integration

The load magnitude is determined through the vehicle weight estimation, and the location of moving vehicles is identified using the vehicle tracking and location estimation, as outlined in Sections 2.2 and 2.3. However, these two pieces of information have not been integrated. Consequently, at this stage of the V-WIM, a framework integration step is performed. This step facilitates the simultaneous identification of the vehicle load and location, thereby enhancing the overall effectiveness of the framework.

Figure 17 illustrates a schematic of the framework integration, encompassing two computer vision-based

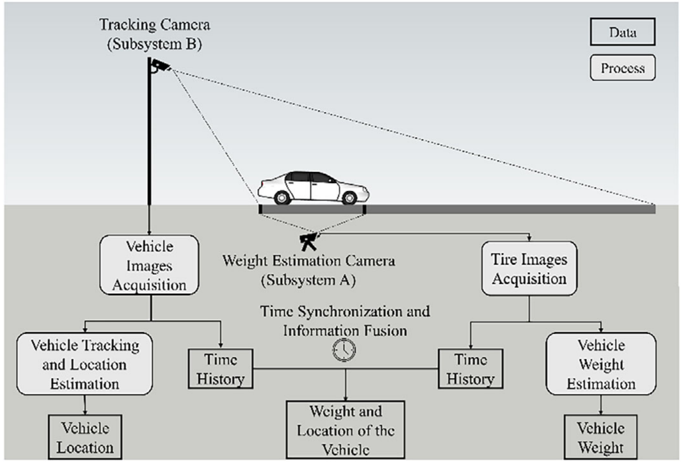

FIGURE 17 Overview of the framework integration.

subsystems. Subsystem A is dedicated to measuring the vehicle load magnitude and recording the time of measurement. Meanwhile, subsystem B is responsible for monitoring the location and time history of the vehicle as it traverses the estimation area. By synchronizing the time data and integrating the information from both subsystems, the framework can simultaneously determine the weight and location of the vehicle.

3 | FRAMEWORK VALIDATION

This section presents a comprehensive validation of the proposed V-WIM framework through a series of tests aimed at assessing its performance. Two separate validation tests were conducted: the first was dedicated to evaluating the vehicle weight estimation, focusing on its accuracy and reliability in determining vehicle weights. The second test examined the vehicle tracking and location estimation, assessing its accuracy in tracking and locating vehicles. Additionally, to confirm the overall effectiveness of the integrated V-WIM framework, a field experiment was conducted. These evaluations provide a robust assessment of each estimation process and the overall framework.

3.1 | Component validation test for vehicle weight estimation

This section evaluates the performance of vehicle weight estimation under various load conditions. First, the tire–road contact length and tire vertical deflection, obtained through computer vision techniques, are validated. Following this, vehicle weights, derived from the validated tire–road contact length and tire vertical deflection, are evaluated. Figure 18 illustrates four load conditions.

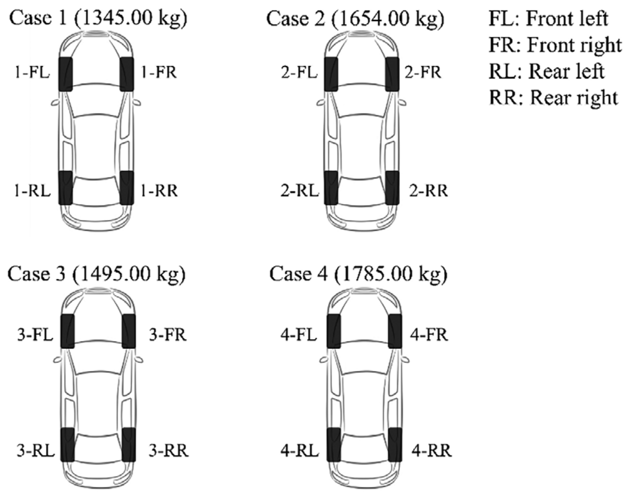

FIGURE 18 Illustration of the four cases used for evaluating the vehicle weight estimation.

3.1.1 | Tire–road contact length and tire vertical deflection estimation

To evaluate the performance of the proposed method for identifying tire–road contact length and tire vertical deflection, a series of experiments was conducted using different vehicles under various cases. The results obtained from the proposed method were then compared with those from ground truth and the manual measurement method. Ground truth was estimated simply using a ruler, while the manual measurement method used in previous works (Feng et al., 2020; Kong, Zhang, et al., 2022) involved quantifying the number of pixels that represent the tire–road contact length and the tire vertical deflection in the image. Due to inherent variability in manual measurements, measurements were performed three times by three different individuals, and the average value was calculated and used as a representative value for this method.

Tables 3 and 4 provide a detailed comparison of the estimated contact length and vertical deflection between

TABLE 3 Comparison of the estimated contact lengths.

			Manual measurement			Proposed method		
				Error			Error	
Case	**Tire**	**Ground truth**	**Contact length**	**cm**	**%**	**Contact length**	**cm**	**%**
Case 1	1-FL	20.50	20.90	0.40	1.95	21.80	1.30	6.34
	1-FR	15.00	13.40	1.60	10.66	16.20	1.20	8.00
	1-RL	13.00	10.80	2.20	16.82	12.20	0.80	6.15
	1-RR	12.50	11.10	1.40	11.20	10.20	2.30	18.40
	Average			1.40	10.15		1.40	9.72
Case 2	2-FL	22.30	19.40	2.90	13.04	23.60	1.30	5.82
	2-FR	16.00	14.10	1.90	11.87	16.50	0.50	3.12
	2-RL	14.30	14.70	0.40	2.79	13.60	0.70	4.89
	2-RR	13.10	11.60	1.50	11.45	13.80	0.70	5.34
	Average			1.67	9.78		0.80	4.79
Case 3	3-FL	15.50	16.30	0.80	5.16	15.90	0.40	2.58
	3-FR	15.50	16.20	0.70	4.51	14.90	0.60	3.87
	3-RL	12.00	13.80	1.80	15.00	13.50	1.50	12.50
	3-RR	12.20	12.70	0.50	4.09	12.10	0.10	0.81
	Average			0.95	7.19		0.65	4.94
Case 4	4-FL	16.00	14.50	1.50	9.37	17.20	1.20	7.50
	4-FR	16.50	15.80	0.70	4.24	17.10	0.60	3.63
	4-RL	15.00	12.80	2.20	14.66	15.90	0.90	6.00
	4-RR	14.50	14.30	0.20	1.37	15.50	1.00	6.89
	Average			1.15	7.41		0.92	5.93
Total average				1.29	8.63		0.94	6.34

the manual measurement method and the proposed method. Table 3 presents the comparison for contact length estimation. Across all four cases examined, the proposed method consistently yields a lower average error than the manual measurement method. Specifically, the manual measurement method results in an average error of 8.63% and a maximum error of 10.15%. Meanwhile, the proposed method achieves an average error of 6.34% and a maximum error of 9.72%, demonstrating its enhanced accuracy in estimating contact length.

Table 4 extends this comparison to vertical deflection estimation. Here, the manual measurement method exhibits an average error of 32.72% and a maximum error of 58.72% over the four cases. The proposed method significantly reduces these error, with an average error of 13.35% and a maximum error of 20.05%. Despite the inherent challenge of estimating vertical deflection due to its small magnitude (measured in millimeters), the proposed method shows promising accuracy, reducing the total average error by 19.37%, compared to the manual method. This demonstrates the effectiveness and reliability of the proposed method in estimating both the contact length and the vertical deflection.

3.1.2 | Vehicle weight estimation

Once the tire–road contact length and tire vertical deflection of vehicles are identified, the weight of the vehicles can be measured. To evaluate the performance of the V-WIM method, four cases with different load conditions were examined as shown in Table 5. The results demonstrated a high degree of accuracy, with all estimated weights recording errors below 7%, except for a single instance that exhibited an error of 11.86%. The minimum error rate achieved was 1.67%, indicating high estimation precision. These findings underscore the robustness of the proposed.

V-WIM framework in reliably determining vehicle weights under a range of loading conditions. The low error rates observed across most cases suggest that the V-WIM framework is highly effective at providing consistent and accurate weight measurements. This robustness is critical for applications requiring precise vehicle weight data, such as transportation safety, infrastructure management, and regulatory compliance. The ability of the V-WIM framework to maintain high accuracy across various loading scenarios highlights its potential for widespread adoption in various applications.

TABLE 4 Comparison of the estimated tire vertical deflections.

			Manual measurement			Proposed method		
				Error			Error	
Case	Tire	Ground truth	Vertical deflection	mm	%	Vertical deflection	mm	%
Case 1	1-FL	18.00	18.90	0.90	5.00	19.10	1.10	6.10
	1-FR	7.00	7.40	0.40	5.71	8.30	1.30	18.57
	1-RL	5.00	4.70	0.30	6.00	5.20	0.20	4.00
	1-RR	2.00	4.80	2.80	140	1.50	0.50	25.00
	Average			1.10	39.17		0.77	13.41
Case 2	2-FL	20.00	16.70	3.30	16.50	20.40	0.40	2.00
	2-FR	10.00	9.10	0.90	9.00	8.80	1.20	12.00
	2-RL	8.00	9.40	1.40	17.50	8.10	0.10	1.25
	2-RR	5.00	5.70	0.70	14.00	5.40	0.40	8.00
	Average			1.57	14.25		0.52	5.81
Case 3	3-FL	9.00	13.70	4.70	52.22	9.60	0.60	6.25
	3-FR	11.00	13.20	2.20	20.00	8.40	2.60	23.63
	3-RL	5.00	9.80	4.80	96.00	6.60	1.60	32.00
	3-RR	6.00	10.00	4.00	66.66	4.90	1.10	18.33
	Average			3.92	58.72		1.47	20.05
Case 4	4-FL	13.0	9.10	3.90	30.00	10.90	2.10	16.15
	4-FR	11.0	10.60	0.40	3.63	10.60	0.40	3.63
	4-RL	11.0	8.10	2.90	26.36	8.60	2.40	21.81
	4-RR	8.0	9.20	1.20	15.00	9.20	1.20	15.00
	Average			2.10	18.74		1.52	14.14
Total average				2.17	32.72		1.07	13.35

TABLE 5 Weight estimation results.

	Weight		Error	
Case	Ground truth	Estimated	kg	%
Case 1	1345.00	1436.41	91.41	6.79
Case 2	1654.00	1850.30	196.30	11.86
Case 3	1495.00	1520.00	25.00	1.67
Case 4	1785.00	1865.77	80.77	4.52

3.2 | Component validation test for vehicle tracking and location estimation

To verify the performance of the deep learning-based vehicle tracking method under outdoor conditions, an experiment was conducted at the Civil Engineering Building at Chungbuk National University. The vehicle moved on a stretch of road with a length of 2100 cm and a width of 350 cm. Additionally, 21 reference points were marked on the surface of the road to facilitate converting the coordinates of the vehicle in image space into real-world space through the coordinate transformation technique. Figure 19a presents the reference points and their positions on the surface of the road.

The moving vehicle was detected by using a pre-trained YOLOv8 detection model. This pre-trained model, trained on the COCO dataset, can detect 80 distinct object categories, encompassing vehicles such as cars, trucks, and buses. In this process, the pixel coordinate of the bottom mid-point in the predicted bounding box of the vehicle in each video frame was converted into the real-world coordinate system (the road surface coordinates system). From that, the position and distance of the vehicle as it moves on the road surface are monitored. Figure 19b(1) illustrates the tracking result. The vehicle was tracked using a detection bounding box, in which its unique ID (obtained from the ByteTrack algorithm) and coordinates on the road surface are displayed. In addition, Figure 19b(2) showed the 2D simulation of the trajectory of the moving vehicle in the tracking process. The reliability of the deep learning-based vehicle tracking method under outdoor conditions was verified by comparing the position of the predicted vehicle to its actual position, affirming the accuracy and robustness of the tracking approach.

3.3 | On-site validation

To ascertain the performance of the integrated V-WIM framework, a field experiment was performed at the Civil Engineering Building, Chungbuk National University.

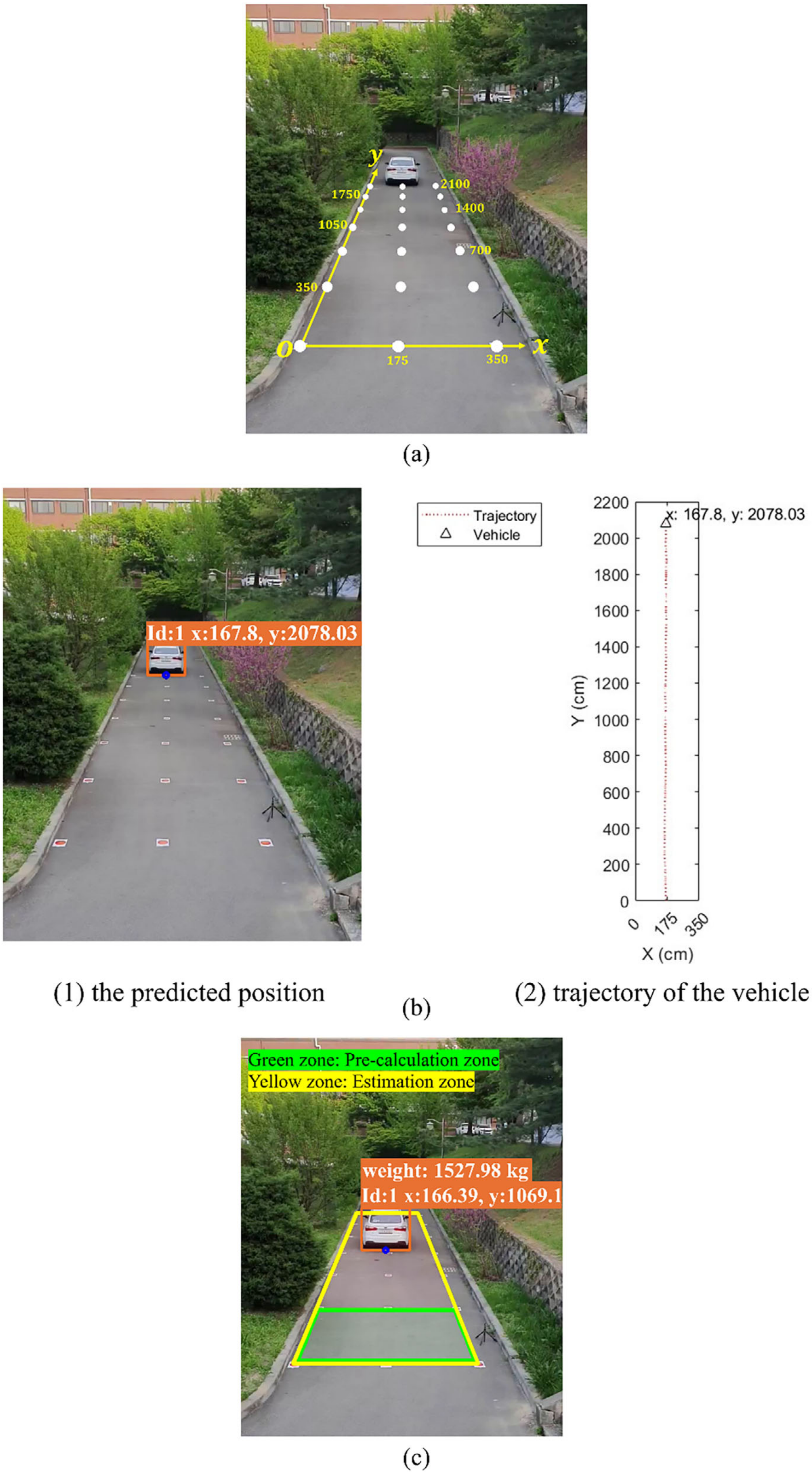

FIGURE 19 (a) Experimental configuration for vehicle tracking under field conditions, (b) vehicle tracking results in outdoor conditions, and (c) field experimental setup for the V-WIM framework.

TABLE 6 Results of on-site test vehicle weight estimation.

Case	Weight		Error	
	Ground truth	Estimated	kg	%
Case 1	1457.00	1527.98	70.98	4.80
Case 2	1706.00	1800.21	94.21	5.50

This experiment was conducted following the integration procedures outlined in Section 2.4, with the objective of simultaneously determining both the load magnitude and position of the vehicle.

An overview of the field test is presented in Figure 19c. Initially, the magnitude of the vehicle load was estimated in the pre-calculation zone (indicated by the green line zone) through the vehicle weight estimation. In this zone, the vehicle moved at a speed of 5km/h, and tire images were captured by an Apple iPhone 8 camera, positioned approximately 100 cm from the vehicle. Concurrently, the position of the vehicle was monitored by the vehicle tracking and location estimation as it entered the estimation zone (indicated by the yellow line zone). By synchronizing the time data and integrating these measurements, both the magnitude of the vehicle load and its location were identified.

A controlled experiment was conducted with a test vehicle. The vehicle load was measured under two specific conditions: with the driver only (Case 1) and with the driver plus four passengers (Case 2). The findings, as summarized in Table 6, include the estimated weights for both scenarios. The weight estimation process exhibited an error of 4.8% in the driver-only scenario and 5.5% in the full-passenger scenario.

Subsequently, the estimated weight of the vehicle, derived from the vehicle weight estimation, was integrated with its location data, which was obtained from the vehicle tracking and location estimation. The process for acquiring the location of the moving vehicle is similar to the procedures described in Section 3.2 of the vehicle tracking and location estimation validation. Figure 19c presents the output of the integrated V-WIM. By synchronizing the measurement data from two estimation processes, this framework effectively identified both the weight and the location of the vehicle, demonstrating the feasibility of using the V-WIM framework for vehicle monitoring within traffic infrastructure systems.

4 | DISCUSSION

The development and validation of the proposed V-WIM framework represents a significant advancement in the field of vehicle load monitoring. By integrating vision-based vehicle weight estimation and vision-based vehicle location estimation into a single framework, this study opens a new path for identifying the load and location of moving vehicles using only visual data, eliminating the reliance on weighing devices or sensors used in existing methods.

The experimental results demonstrate that the proposed vision-based method is an effective approach for estimating tire deformation parameters and vehicle weight. Regarding tire deformation parameters estimation, this study employs a detection model (YOLOv8) to detect the tire, rim, and sidewall text components from tire images and leverages their bounding boxes to extract tire deformation parameters. This approach is straightforward and efficient, compared to image processing-based or image segmentation-based methods, which are often considered manual process, prone to high variability, or laborious, as seen in previous works.

Specifically, in contact length estimation, compared to the manual measurement method, which exhibits an average error of 8.63% and a maximum error of 10.15%, the proposed method achieves an average error of 6.34% and a maximum error of 9.72%, demonstrating improved accuracy. Furthermore, in tire vertical deflection estimation, despite the inherent challenge of accurately estimating small values (measured in millimeters), the proposed method achieves an average error of 13.35% and a maximum error of 20.05%, while the manual method displays a substantial average error of 32.72% with a maximum error of 58.72%. The proposed method shows promising accuracy, reducing the total average error by 19.37%, compared to the manual method.

Building on these improvements, vehicle weight estimation also achieves significant improvements in all experimental cases. For four cases in the component validation test, estimated weights recorded errors below 7%, except for a single case that exhibited an error of 11.86%. The minimum error rate achieved was 1.67%. For the on-site test, the estimated weights showed errors of 4.8% and 5.5% for each case.

Regarding vehicle tracking and location estimation, while previous studies have primarily focused on vehicle detection, our proposed tracking strategy, which integrates YOLOv8 and ByteTrack, allows for accurate and consistent tracking of moving vehicles by assigning a unique ID to each vehicle. Assigning IDs to vehicles is a key feature, providing essential data for vehicle monitoring and enhancing its effectiveness.

Although the proposed framework demonstrates reasonable accuracy in most experimental cases, significant errors can occur due to current limitations, particularly evident in Case 2 of the component validation test, where an error of 11.86% (corresponding to 196.30 kg) was observed. The rest of this section discusses limitations and explores potential mitigation strategies.

4.1 | Limitation of dataset

Although the proposed method performs well with the current dataset, the diversity of vehicle types in real-world conditions could pose a challenge in maintaining the high accuracy of the framework. Therefore, continuously adding tire images from diverse types of vehicles can enhance the generalization ability of the detection model when operating with various vehicle types.

4.2 | Limitation of dependence on accurate predicted bounding boxes

Estimating the tire–road contact area relies on the predicted bounding boxes. Inaccurate bounding boxes can introduce errors into the estimation process. To address this issue, an aggregating results strategy can be implemented. Multiple tire images from different frames can be processed simultaneously. By aggregating results from multiple frames, the impact of individual inaccuracies in bounding box predictions can be minimized, thereby ensuring overall accuracy. Additionally, post-processing steps, such as incorporating geometric constraints and applying circle detection techniques, can further refine the predicted bounding boxes to improve alignment with the actual tire shape.

4.3 | Limitation of tire pressure

In real-world scenarios, tire pressure can vary, which might influence tire deformation and consequently affect the accuracy of vehicle weight estimation. In recent years, installing tire pressure monitoring systems (TPMS) on vehicles has become increasingly common. Notably, commercial trucks manufactured after 2012 are mandated to have TPMS installed (Ogunwemimo, 2011). Moreover, advancements in TPMS technology now enable the wireless acquisition of accurate tire pressure data (Vasanthara & Krishnamoorthy, 2016). Wirelessly acquiring accurate tire pressure is a promising strategy to address the limitations of our current approach.

4.4 | Limitation of environmental conditions

Various weather conditions such as fog, snow, and rain are also factors that can affect the performance of the framework. Even though our dataset includes images captured under snowfall or light rain, the number of these images are fewer than that of images captured in good conditions due to the infrequent occurrence of snow or rain days. To overcome this problem, several data augmentation techniques, including blurring, adding noise, and adjusting brightness, were applied to our dataset to simulate how images might appear under various weather conditions. In future work, state-of-the-art domain adaptation object detection methods (Du et al., 2024) could be employed to enable the detection model to perform effectively in different conditions without requiring images captured under those conditions.

Other factors, such as road roughness or vehicle speeds, should be considered. Road roughness can affect the deformation of the tires. However, the road surface where current in-motion vehicle weighing methods (P-WIM, B-WIM) are deployed is typically well-maintained to ensure accurate weighing results. Therefore, it is reasonable to assume that the proposed framework would also be deployed on a well-maintained road upon deployment. Regarding vehicle speeds, the experiments were conducted with vehicles moving at low speeds (5 km/h) to ensure the capture of high-quality tire images. However, in real-world traffic scenarios, vehicle speeds may be higher, making it challenging to acquire clear tire images. This challenge can be addressed by installing specialized cameras capable of capturing high-resolution images of fast-moving vehicles.

5 | CONCLUSION

This study proposes a V-WIM framework that contributes to providing a straightforward yet efficient vehicle load identification. The main methodological contribution of this study is that this is the first effort in integrating vision-based vehicle weight estimation and vision-based vehicle location estimation into a single framework. As a result, the load and location of moving vehicles can be identified using only visual data, eliminating the dependence on weighing devices or sensors. Both the component validation test and on-site validation test confirmed the feasibility and effectiveness of the proposed method. The following conclusions can be drawn:

1. The methodology of computer vision-based vehicle weighing was developed. First, an object detection algorithm was employed to detect tire, rim, and sidewall text from tire images. Then, the OCR algorithm was used for the tire specifications recognition from the detected sidewall text. Moreover, the detected bounding boxes of the tire and rim were used to estimate the tire deformation parameters such as tire–road contact length, tire

vertical deflection. From that, the relationship of the tire–road contact area, the tire–road contact pressure, and the tire–road contact force were used to determine vehicle weight.

2. The integration of YOLOv8 and ByteTrack allows for robust and consistent tracking of moving vehicles by assigning a unique ID to each vehicle. Assigning IDs to vehicles is a key feature, providing crucial information for monitoring vehicles, which pushes vehicle monitoring to a higher level.
3. A field test was conducted to validate the performance of the integrated framework. The results show that the experimental vehicle weights were estimated with reasonable accuracy, while the location of the experimental vehicle was simultaneously tracked during the testing process.
4. Future research should focus on enhancing the vehicle weight estimation to ensure reliable performance under real-world traffic conditions. Self-supervised learning techniques offer significant potential to improve generalization and robustness of the proposed method (Rafiei et al., 2024a, 2024b), particularly when faced with diverse vehicle types and environmental factors. Exploring these advanced techniques represents a promising direction for improving the current research.

Funding information

This work was upported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2022-NR071167 and RS-2024-00454760)

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How to cite this article: Lam, P. T., Lee, J., Lee, Y., Nguyen, X. T., Vy, V., Han, K., & Yoon, H. (2025). A vision-based weigh-in-motion approach for vehicle load tracking and identification. *Computer-Aided Civil and Infrastructure Engineering*, *40*, 1466–1489. https://doi.org/10.1111/mice.13461